CS430/910: Foundations of Data Analytics

Case Studies| Dr Ayse Sunar

Objectives

- Going through the details of real-world applied data analytics project
- Understand the logic of the processes involved in data analytics
- See visualisation examples and their interpretations

Have you decided your data for project?

- Have you chosen your data?
- Have you identified research questions?
- Have you chosen your tools?
  - Python, Google Colab, Jupyter lab etc.?

Case Studies

- 4 papers on data analytics published in the scientific literature:
  - I Tube, You Tube, Everybody Tubes: Analyzing the World's Largest User Generated Content Video System
    Internet Measurement Conference 2007
  - What is Twitter, a Social Network or a News Media?
    19th international conference on World wide web 2010
  - Meme-tracking and the Dynamics of the News Cycle
    Knowledge Discovery and Data Mining (KDD), 2009
  - Link Prediction by De-anonymization: How We Won the Kaggle Social Network Challenge
    2011 International Joint Conference on Neural Networks (IJCNN)
- Read them! – recommendation

Details on Case Studies

- Please read the papers in detail to get the full story
- Bias: papers are from Computer Science research community
  - Mostly address data analysis applied to large websites
  - The most well-studied example of “Big Data”
  - Examples should be familiar to you (YouTube, Facebook, Twitter)
- Objectives for the case studies:
  - To see examples of data analytics in practice
  - To introduce and motivate topics we will study in more detail later
  - To see examples of going from data to insight to understanding

Case Study 1: Online video

- "I Tube, You Tube, Everybody Tubes: Analyzing the world's largest user generated content video system"
- By Meeyoung Cha, Haewoon Kwak, Pablo Rodriguez, Yong-Yeol Ahn, Sue Moon (Telefonica Research and KAIST)
- Published in Internet Measurement Conference 2007
- http://conferences.sigcomm.org/imc/2007/papers/imc131.pdf

Objectives of the paper

- To understand the impact of video sharing systems
- To study the popularity life-cycle of videos
- Study statistical properties of requests, relation to video age
- [Study prevalence of copying activities]
- Understand potential for caching to save bandwidth

Data Collection

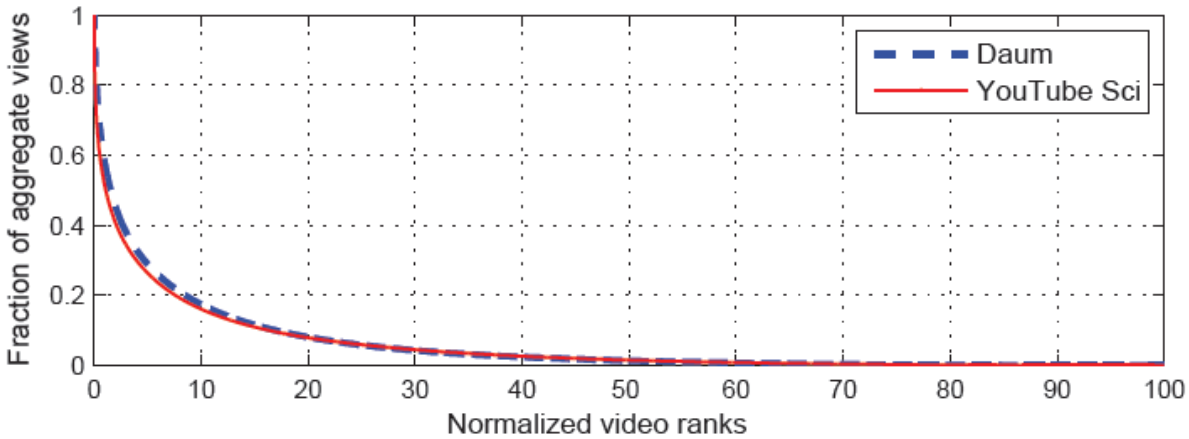
Name	Category	# Videos	Tot. views	Tot. length	Data collection period
YouTube	Ent	1,687,506	3,708,600,000	15.2 years	Dec 28, 2006 (crawled once)
YouTube	Sci	252,255	539,868,316	1.8 years	Jan 14 - 19, '07 (daily), Feb 14, '07, Mar 15, '07 (once)
Daum	All	196,037	207,555,622	1.0 year	Mar 1, 2007 (crawled once)

- Crawled YouTube and Daum sites in 2007
  - Wrote programs to automatically collect data about all videos
- YouTube was already very large in 2007
  - Restricted crawl to 'Entertainment' and 'Science/Tech' categories
- Collected data on each video:
  - Fixed: Uploader id, date of upload, duration of video
  - Variable: #views, #total ratings, #positive ratings, links to
- Daily crawl for 6 days to see changes

Video popularity distribution

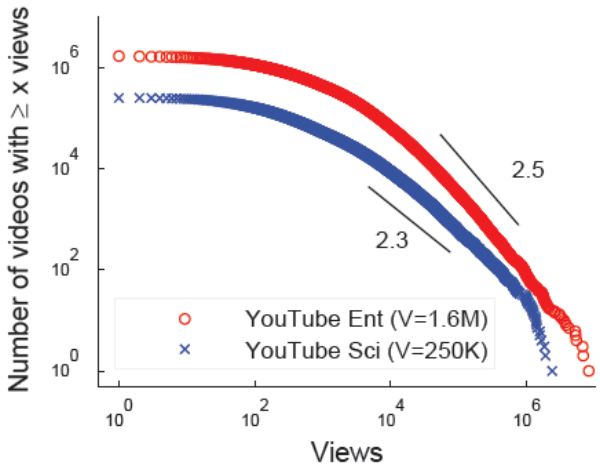
- Plot what fraction of views are **outside** the top videos
- Normalise ranks from 0 to 100, to allow comparison
- Top 10% of videos account for 80% of views
  - Very skewed distribution
  - Wide variation in popularity. Why?

Scaling and normalising the variables are very important to compare the results!



Understanding video popularity distribution

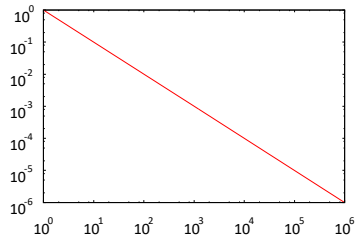
- “Skewness” (“the long tail”) is a common phenomena in data
- Observed by plotting data on a log-log scale: straight lines (because a few videos have many views, and most have far fewer.)
- Plot views on x-axis, #videos with more than x views on y-axis



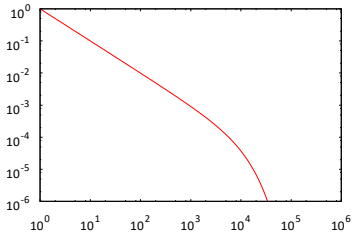
Modeling Skewness

Several distributions generate skew

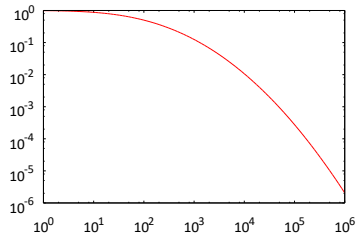
- "Power law" (Pareto, zipf): y proportional to $x^{-\alpha}$ for some α
  - Gives straight line on log-log plot
- "Power law with exponential cut-off": y proportional to $x^{-\alpha}e^{-\beta x}$
  - $x<1/\beta$: behaves like power law
  - $x>1/\beta$: behaves as exponential decay
- Log-normal: taking log of distribution produces a Normal curve



A power law describes a relationship where one quantity varies as a power of another. In this case, as *x* increases, *y* decreases rapidly.



This distribution behaves like a power law for smaller values of *x*, but for larger values of *x*, the exponential cutoff dominates, leading to a rapid decay in *y*.

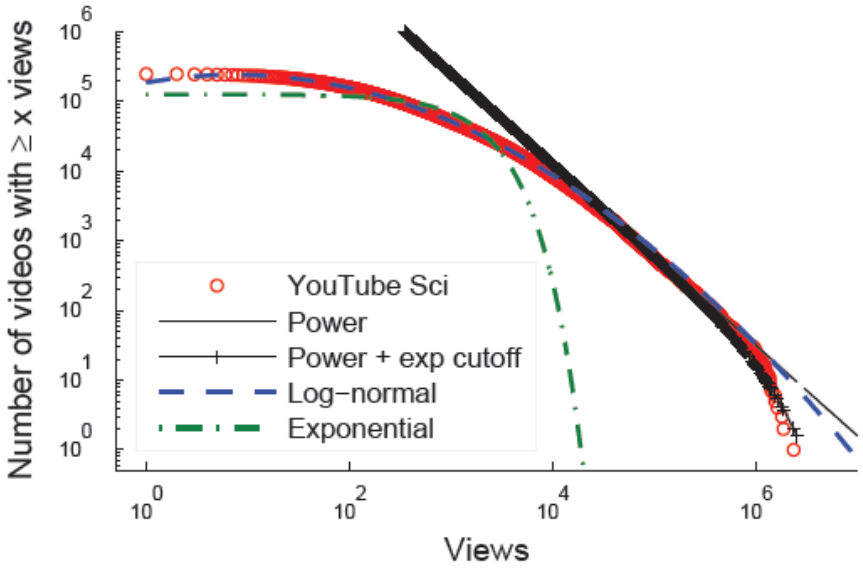


A log-normal distribution is one where taking the logarithm of the data transforms it into a normal (bell-shaped) distribution.

How to tell which we are seeing?

Fit a curve

- Find the best fitting curve for each model
  - A regression problem (covered later)
  - Log-normal captures behaviour for popular videos (head)
  - Power law with exponential cutoff seems best for tail
- Why?



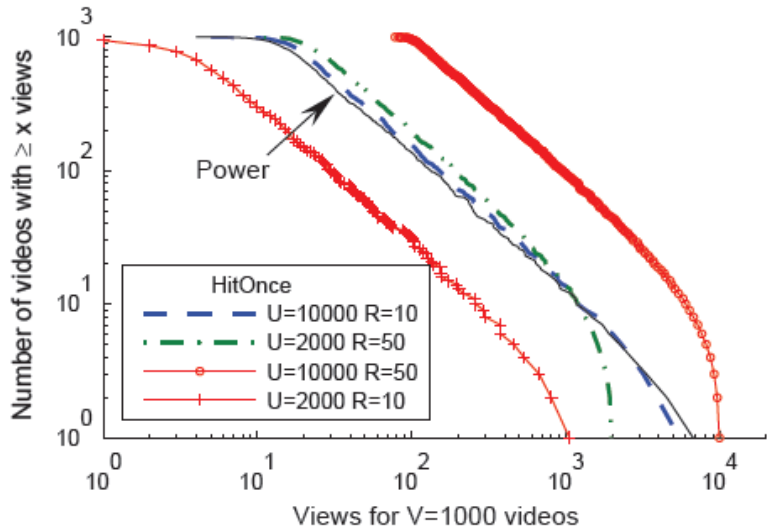
Possible explanations

Several mechanisms are known to generate long tail distributions:

- Preferential attachment: popular items are most likely
  - Describes the main behaviour, but not the truncated tail
- Aging effect: “old” items eventually die, receive no new activity
  - Does not fit videos: no ‘death’ or ‘removal’ of an old video
- Information filtering: a user can only view a fixed number
  - Does not fit: people can keep watching new videos
- Fetch-at-most-once: a user can view each video at most once
  - Better: people prefer to watch new videos, don’t watch top-10 over and over
    - Some exceptions: music fans watch favourites many times

Validating ‘fetch-at-most-once’

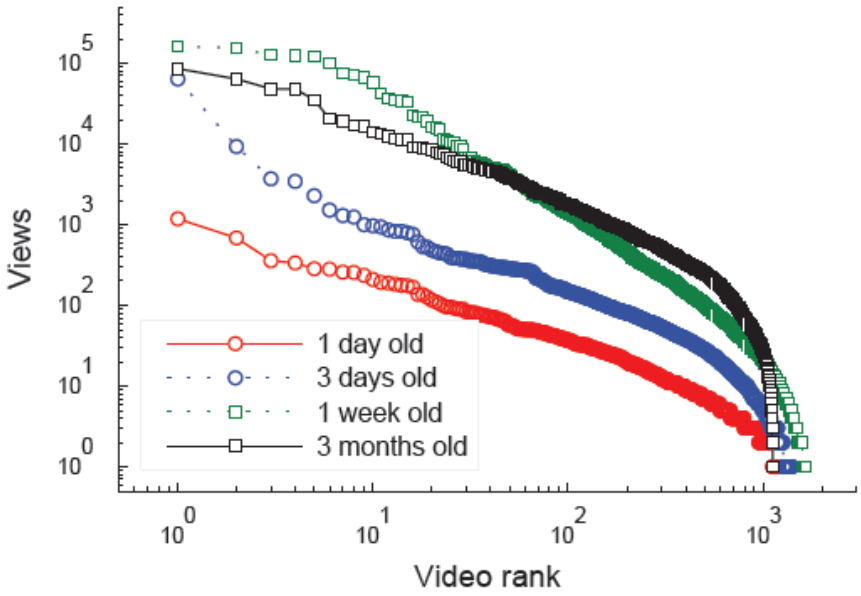
- Simulate preferential attachment + fetch-at-most-once
  - R requests per user
  - U different users
  - V different videos
- Observations from simulation
  - Increasing R sharpens tail
  - Increasing U shifts graph
    - Shape doesn’t change much



- Do you agree?

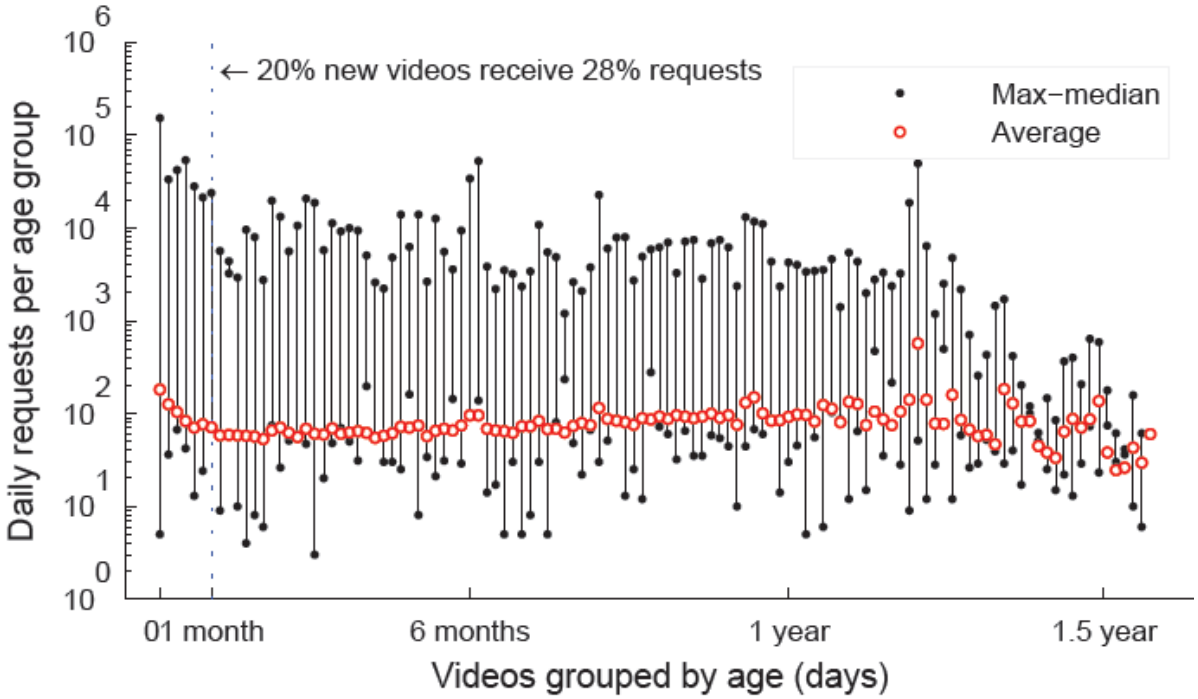
Effect of time

- Views increase over time
- Truncation gets sharper over time
  - Many possible reasons: e.g. more push to most popular content



Impact of age (of video)

- Popularity does not vary strongly by age
- Most recent videos are slightly more popular
- Data is from early days in YouTube: have things changed?



Possible new questions:
-Have things changed? Towards which trend? Possible societal reasons to explain?

- Tiktok as a popular video sharing site has the similar behaviour? Why or why not?

Can we predict future popularity?

- Consider current popularity (views) and age (time since upload)
  - Does this correlate with future popularity?

Age (x_0)	x_0+5 days old	7 days old	90 days old
2 days old	0.9665 (5185)	0.8793 (3394)	0.8425 (11215)
3 days old	0.9367 (3394)	0.9367 (3394)	0.8525 (9816)

  - Table shows correlation coefficient (number of videos sampled)

- Strong correlation of instant popularity with future popularity
  - From day 2. Day 3 does not change much.

Use these observations to cache

Type	Size	# missed videos	# missed requests
Static	41,235	115,002 (48.8%)	5,093,832 (26.7%)
Dynamic	256,647	4,683 (1.9%)	648,376 (3.4%)
Hybrid	51,235	94,893 (40.3%)	3,271,649 (17.1%)

- Streaming video uses a lot of Internet bandwidth (up to 66%?)
- Could we cut bandwidth usage by running a cache?
  - E.g. put a video cache for all of Warwick University
- How to fill the cache?
  - Static: Pick the most popular items once and for all
  - Dynamic: Initialize with most popular, then cache all new videos
    - Unrealistic, but a point of comparison
  - Hybrid: Static + daily most popular

CONTEXT:

A potential solution to a large amount of bandwidth by streaming video is running a **video cache**, where frequently accessed videos are stored locally (on a server close to users) to minimise repeated requests for the same video over the Internet.

Reflections on the paper

- A widely referenced paper from early in YouTube's history
  - 1100+ citations in the literature
- Characterised many aspects of video viewing behaviour
  - And attempted to explain many of these
  - Many other plots in the paper
- Video on the Internet has changed a lot since 2007
  - Changes to YouTube website structure
  - Huge growth in mobile devices
  - Videos with billions of views
- Do conclusions still hold? What other phenomena emerge?

Case Study 2: Microblogging

RIP Twitter lives in scientific papers and lectures.

- "What is Twitter, a social network or a news media?"
  - Kwak, Lee, Park, Moon (KAIST), in WWW conference 2010
  - https://dl.acm.org/doi/10.1145/1772690.1772751

  Objectives:

  - "To study the topological characteristics of Twitter and its power as a new medium of information sharing"
  - "The first quantitative study on the entire Twittersphere and information diffusion on it"

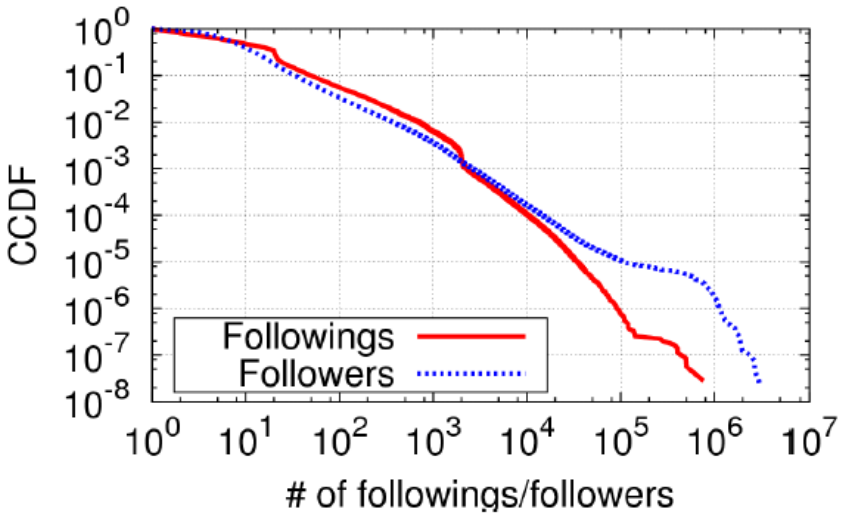
One slide on Twitter

- Messaging service where messages are up to 140 characters
- Broadcast (by default) to all other users
  - @userid addresses a particular user
  - #hashtag to tag a message
  - RT: re-tweet someone else's message (sometimes with comment)
- Users "follow" another, and receive that user's messages

Data Collection in “What is Twitter”

- User Profiles
  - Crawl name, location, timezone, number of tweets
  - Begin at a popular user to crawl the “giant connected component”
  - Twitter used to limit to 20,000 requests an hour
    - Using 20 machines with different IP addresses, took 24 days
- Tweets
  - Collected all tweets mentioning current “trending topics”
    - Probed every 5 minutes
  - Used Twitter search API to get up to 1500 tweets per query
  - Collect text, author, timestamp
  - Removed “spam tweets” from very new users

Follower/following analysis

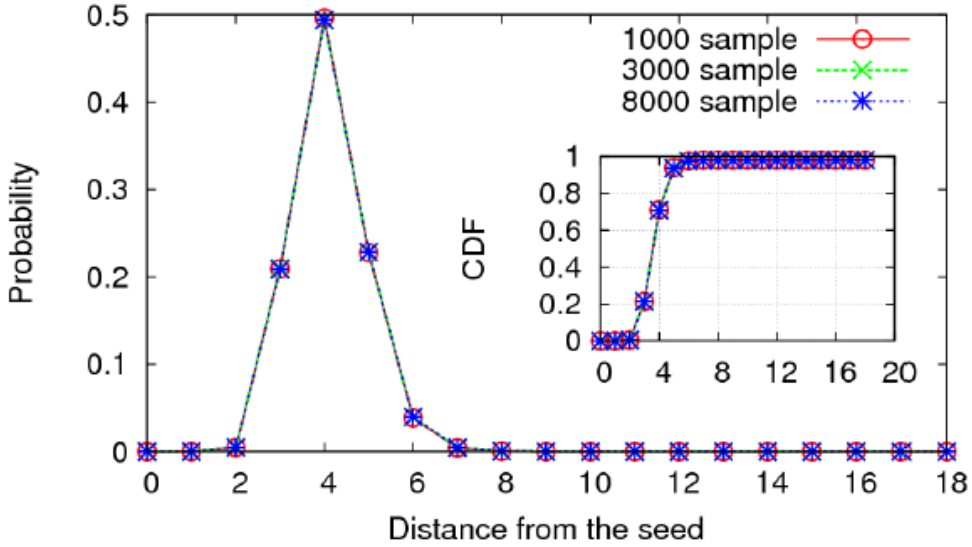


CCDF (Complementary Cumulative Distribution Function):
Fraction with more than this number

- Asymmetry between followers and following
- Some bumps can be explained with domain knowledge
  - Following 20: Twitter suggests an initial set of 20 to follow
  - Following 2000: used to be a limit of 2000, since removed
- Fits power law, with exponent 2.28 (quite skewed)

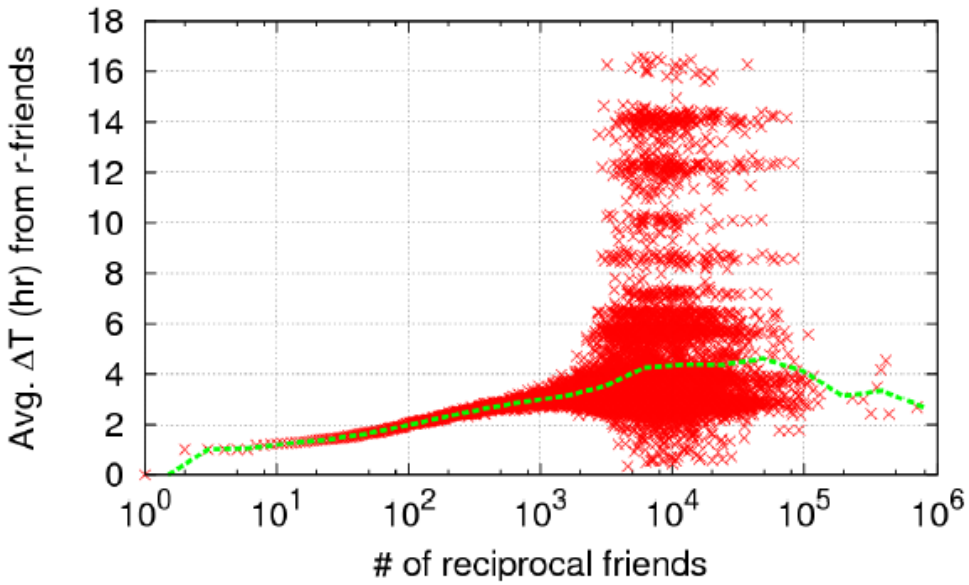
Degree of separation

- 2 users are “friends” if there is a mutual following relationship
- Study the friendship distance from random starting points
  - 80% are within 4 steps (compare to “6 degrees of separation”)



Proximity of users

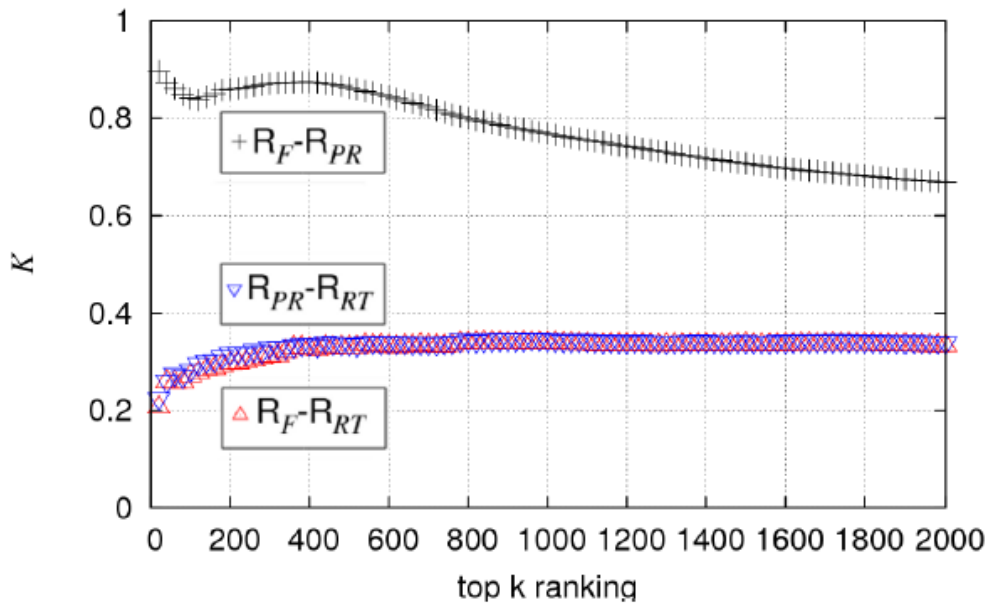
- Are friends geographically close? How would you test this?
- Look as average time difference between friends
  - As a function of number of friends



Importance of Twitter users

- Who are most important/influential Twitter users?
  - F: Count followers? Too crude?
  - R: Most retweeted?
  - PR: Most critical in the follower-following graph?
    - Use PageRank: defined measure importance of web pages
    - Recursive definition: PageRank of a node is sum of PageRanks of its followers
    - Can be computed efficiently (see later)
- Compare top-20 in each: they appear similar
- Compute rank-correlation between top-k

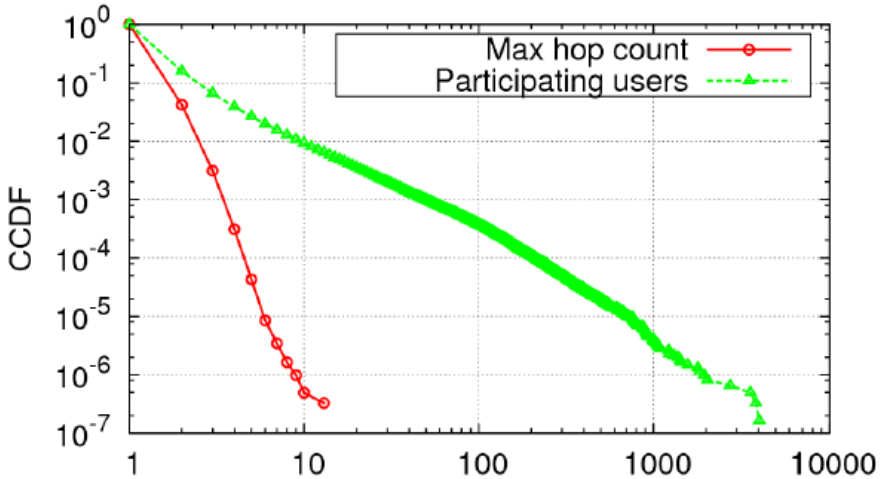
Comparison of importance measures

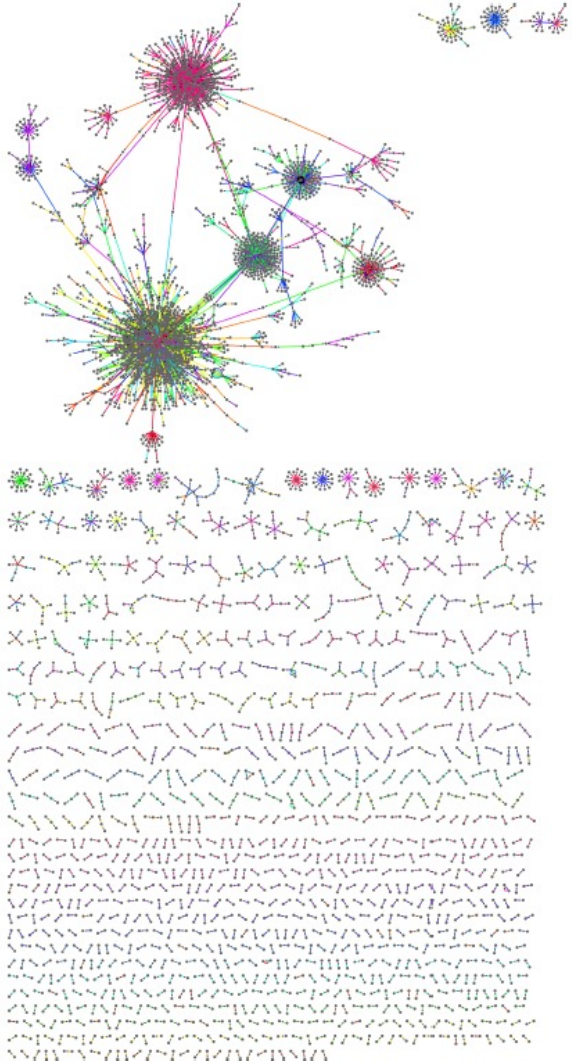


- Followers (F) and PageRank (PR) give most similar ranking
- ReTweets (RT) are also correlated, but more weakly so

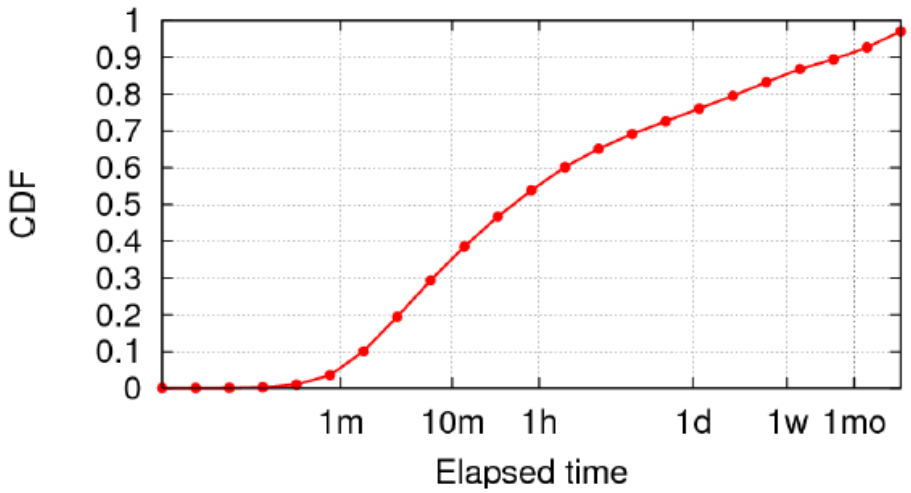
Spread of influence

- ReTweets can spread a message far and wide
- Figure shows retweets of messages about a plane crash
  - Most trees are shallow (<3 hops)



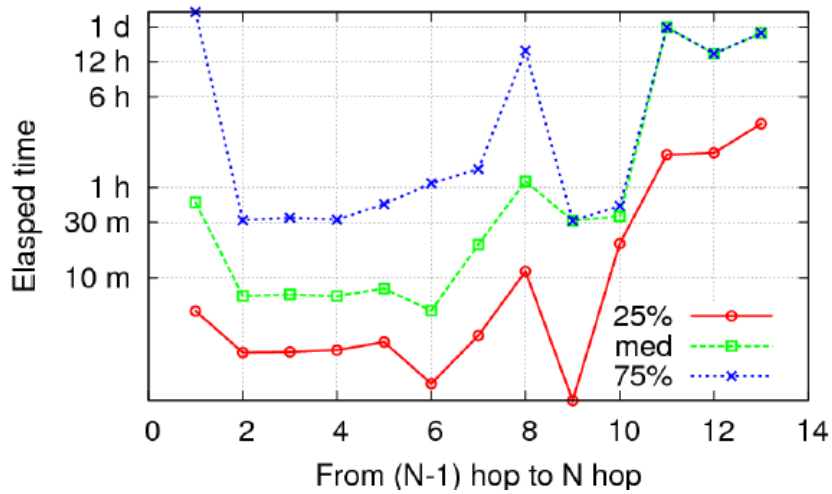


Retweet time



Distribution of delay from initial tweet to retweet

Inter-hop delay through retweet trees



- What do these plots tell you?

Reflections on the paper

- See also "A few chirps about twitter", Krisnamurthy, Gill, Arlitt
  - From 2008, early in Twitter's history
  - Instructive to compare the approach and the findings
- Shows that there are many ways to slice and dice data from a relatively "simple" data source
  - Did not even look much at content of tweets
- Widely cited (3400+ citations) as early work on Twitter

Acknowledgements

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  - G. Cormode [Warwick, CS910]
  - F. Ciucu [Warwick, CS430/CS910]

Questions?